

Hanoon Prime — Architecture Study

Date: 2026-09-23 · **Repo:** `/Users/mdsabersajib/Downloads/hanoon_prime` (user's Mac, read-only study — nothing modified) **Method:** Four parallel code-study workstreams, each reading the Mac repo via `files.read / files.list` only. Every technical claim below is cited as `repo-relative-path:line` and was verified by direct read on 2026-09-23. Exhaustive detail lives in the part files: `part1_live_path.md`, `part2_organs.md`, `part3_safety.md`, `part4_validation.md` (same directory).

One-line summary: a ~29k-line, 185-module JULI-brain → IB-Gateway paper-trading system on IBKR paper (port 4002) whose docs are unusually honest — its own walk-forward gate currently FAILS — and whose live path contains several verified safety and scoring defects detailed in §8.

1. Live money path: tick → order fill

1.1 Boot (once)

1. `python3 -m hanoon_prime.cli → main()` — `src/hanoon_prime/cli.py:85-114`. Tickers = CLI args, else seed universe `LIQUID_US_SEED` (22 names) — `cli.py:94`, `immune.py:227-249`.
2. `IBStreamingBot(account="PAPER")` — `cli.py:98` → `ib_adapter.py:48-101`. Wires `JuliBrain`, `IBStreamer`, `IBExecutor`, `PipelineMonitor`; journal → `runtime/journal_live.jsonl`; `executor.on_fill_confirmed = self._confirm_fill`. Note the bot's own `Hippocampus(safety_enabled=False)` — `ib_adapter.py:64`.
3. `bot.run_paper(tickers)` → `connect(port=IB_PAPER_PORT=4002)` → `run()` — `ib_adapter.py:197-200`; main loop `while self._running: self._cycle(poll, pnl)` — `ib_adapter.py:160-180`.

1.2 The cycle (~1 s)

`BotCycleMixin._cycle` — `ib_cycle.py:487-518`, in order:

1. Sleep-manager check; gateway supervision (reconnect + resubscribe).
2. `executor.sync_from_ib(...)` — IB is source of truth: sweep zombies, adopt orphans, re-attach protection (`_protect.py`), rebuild `_brackets` (`ib_bracket.py:29-47`), `read_ib_positions` (`_ib_sync.py:106-134`).
3. `_sweep_stale_orders()` — cancels stale (>60 s, PendingSubmit/PreSubmitted) DAY entry parents — `ib_cycle.py:262-298`.
4. `_sync_subs()` — market-data subscriptions from the rotated budget pool.
5. Manual/EOD flatten (`ib_cycle.py:845-906`) — **skips the brain**, sends MKT flattens directly.
6. `_run_brain_cycle(poll, pnl, started)` — `ib_cycle.py:561-598` → `juli.tick(watch, ...)` — `juli.py:44-74`.

JuliBrain.tick : re-entrancy lock → governor.begin_cycle() (resets 2-entry budget — orchestrator.py:258-260, policy/governor.py:26-28) → _data_preamble → _eval_window (juli.py:76-106; tickers with ≥20 prices, READY_FLOOR=20, juli.py:21; no-data rest rotated 4/cycle, EVAL_WINDOW=4, juli.py:20) → each ticker brain.decide_entry(...) → _evaluate_exits (juli.py:145-175).

1.3 Entry decision — decide_entry (brain/orchestrator.py:263-306), exact order

1. _check_snapshot_valid — veto no_data if snap missing / <20 prices / NaN last/bid/ask/mid — orchestrator.py:318-330.
2. _score_candidate (orchestrator.py:336-366): compute_alpha_from_snap — juli_feed.py:59-77 → compute_all_alpha (**5 core + 22 higher-order = 27 keys** — brain/indicators.py:30-55), fallback to 5-core cerebellum.compute_alpha, plus live tape absorption keys.
3. self.tick → _evaluate_fast (orchestrator.py:837-873): regime fallback → cross-asset modifier (clamped ± CROSS_ASSET_MOD_BOUND=0.04) → horizon classify → horizon bandit → absorption-scalp override → regime weight hot-swap → strategy select → episodic bias (± EPISODIC_MOD_BOUND=0.10) → **_score_pipeline**.
4. **_score_pipeline** (orchestrator.py:1286-1319):
 - cortex._threshold = dynamics.threshold; base = cortex.evaluate(alpha) — Z-score → $\tanh(\sum w \cdot z / \sum |w|)$ over 27 keys (cortex.py:80-115).
 - nash_pred modifier (orchestrator.py:770-772); neuro_score = **always 0.0** (NEURO_BLEND_ENABLED=False — immune.py:132, orchestrator.py:747-768).
 - blended = $0.7 \times (\text{base.score} + \text{cal_adj}) + 0.3 \times 0$ — **a 30% damping of cortex, not a live blend** (orchestrator.py:1296).
 - mods = halim (±0.03) + episodic + nash_op + news (±0.03) + cross – advisor_delta + thinker + strategy (orchestrator.py:1233-1253); raw = blended.regime_mul + somatic + precision.mods.
 - _stabilize (orchestrator.py:1386-1397): Nash bounded penalty (NASH_PENALTY_MAX=0.15, veto bands 0.45/0.55 — brain/config.py:196-199) → EOD zero → dynamics.process (hysteresis ±0.03).
5. _maybe_size (orchestrator.py:1399-1422): if |stabilized| ≤ dynamics.threshold × patience (scalp patience 1.00 — horizons.py:58-65) → HOLD not_sized. Else risk.evaluate(...) (brain/risk.py:190-252): preflight, Kelly × KELLY_FRACTION=0.25, realized-EV (**advisory only** — _ev_scale ∈ {1.0, 0.75, 0.5}, "NEVER refuses" — risk.py:36-41, 190-205); open_positions ≥ 3 → shares=0; quality penalties (VWAP chase ≤0.10, momentum ≤0.08, spread ≤0.06 — brain/config.py:91-103); _size (risk.py:107-145): shares = max(1, min(5000/price, 50/risk_per_share, ...)) ; stop 2.0×ATR, target 6.0×ATR (3:1).
6. _extract_thought : direction≠0 and |score| ≥ DIRECTION_MIN_SCORE=0.02 else HOLD no_signal (immune.py:60, orchestrator.py:368-388).
7. _apply_fast_gates (orchestrator.py:419-448): session → direction → penny → halt (+ safety authorized snapshot).

8. `_apply_strategy_gates` (`orchestrator.py:449-483`): `ENTRY_COST_AVERSE_GATE` / `ENTRY_REGIME_GATE` both False → pass-through.
9. `governor.may_enter` : ≤ 2 entries/cycle (`immune.py:55`), 60 s per-ticker cooldown (`immune.py:56`) — `policy/governor.py:31-42` .
10. `_admit_verdict` (`orchestrator.py:535-552`): `_dnn_gate_veto` (`META_DNN_ENABLED=True` , `META_WIN_THRESHOLD=0.52` — `brain/learning_config.py:34-36`) → `_portfolio_admit` (`orchestrator.py:587-625` ; `exposure<1.0`, `concentration $\leq$ 0.25`, `count<3` — `policy/portfolio_gate.py:55-96`) → ENTER verdict.

1.4 Verdict → order

`_finish_cycle` (`ib_cycle.py:600-633`) → `_execute_entries` (`ib_cycle.py:645-662`): ENTER only if `market_open` **and** `monitor.bar_feed_fresh()` , else "ENTRY SUPPRESSED: bar feed stale" → `_execute_verdict` (`ib_cycle.py:690-716`): `sizing>0`, live bid/ask, not already open → `price = mid` → `executor.place_bracket(...)` → `brain.note_entry` (cooldown stamped **at placement** — `ib_cycle.py:716`). `place_bracket` (`ib_executor.py:62-151`): `ib.bracketOrder(action, shares, round(mid,2), target, stop)` , `tif="DAY"` , `outsideRth=True` — **parent is a LIMIT at the rounded mid** (`ib_insync` builds a `LimitOrder` parent); children = LMT take-profit + STP stop-loss (OCA). SELL blocked under `long_only` . Fill accounting: `sync_from_ib` → `_notify_open_fills` (`ib_executor.py:403-418`) → `_confirm_fill` (`ib_cycle.py:928-941`): `brain.register_position` → `ExitPolicy.register` (`brain/exits.py:329-346`); `_attach_position_watchers` (`ib_cycle.py:943-982`) enqueues `hard_stop_breach` on tick; `_drain_event_exits` (`ib_cycle.py:984-997`) closes sub-cycle.

1.5 Exit path (how positions close)

1. **Brain exits** — `juli._evaluate_exits` → `brain.check_exit` (`orchestrator.py:1711-1739`) → `ExitLadder.evaluate` (`brain/exit_ladder.py:62-104`), first decisive tier wins:
 - TIER1 hard stop: price breaches `stop_price` — absolute (`exit_ladder.py:139-153`).
 - TIER2 JULI verdict: `ExitPolicy.exit_likelihood` (8 pillars — `brain/exits.py:283-327`) vs learned thresholds (`adaptive_thresholds.json` , 2,091 updates); hysteresis disabled (`HYSTERESIS_EXIT_ENABLED=False` — `immune.py:128`) → soft exits immediate.
 - TIER3 mechanical `ExitPolicy.evaluate` (`brain/exits.py:348-390`): absorption-break → `profit_lock` tiers (`brain/config.py:77-82`) → `giveback` (`GIVEBACK_KEEP_RATIO=0.55`) → stale (`STALE_EXIT_MINUTES=120.0`) → consolidation ($<0.1\%$ for 6 pulses).
 - → `close_position` — **MKT DAY** `outsideRth` (`ib_executor.py:531-552`). IB-native bracket children also fire independently.
2. **Tick-watcher hard stop** — sub-cycle MKT close on breach.
3. **EOD flatten** — last 5 min (`policy/trading_policy.py:41-42`); intraday horizons only (`holds_through_close` — `horizons.py:110-113`); brain skipped.
4. Close → `_record_exit` (`ib_executor.py:285-330`): P&L via `get_ib_pnl` (IB fills → return fraction — `_ib_sync.py:19-27`); journaled; appended to `_closed_trades` .

1.6 Learning loop (per close)

`_reflect_closed` (`ib_cycle.py:1005-1033`) → `brain.on_trade_close` (`orchestrator.py:1450-1500`):

1. `_ironclade_gate` — only sources `{"real_trade", "ib_fill", "ib_paper", "reconciled_exit"}`
`learn` (`brain/config.py:209-214`).
2. `_adapt_threshold` → `dynamics.adapt_threshold` (+0.01 if `err` ≥ 0.50, -0.005 if < 0.45; clamped [0.45, 0.70] — `brain/dynamics.py:112-143`, `brain/config.py:29-31`); `memory.threshold` persists.
3. `_update_rpe`; `_learn_exit_thresholds` (`AdaptiveThresholds.update_from_outcome`);
`exits.deregister`; `nash.record_outcome`; `_store_neuromorphic_outcome` (STDP traces mutate, write-back gated); `episodic.add`; `extinction.record`; `_learn_strategy_organs` (meta-label, horizon bandit, strategy bandit/registry; `REGIME_MIN_TRADES=15`).
4. `_learn_from_real` (`orchestrator.py:1570-1616`): `Reflector.on_trade_close` — loss-aversion 2.4:1 (`PENALTY_SCALE=1.2` vs `REWARD_SCALE=0.5`, LR 0.02, decay 0.999 — `brain/config.py:52-58`)
→ regime-weight hot-swap → `realized.add_outcome` → `exits.adapt_from_realized` (retunes giveback keep-ratio ±0.20, stale window ±60 min — `brain/exits.py:254-281`, `brain/config.py:84-88`).

1.7 Gate/veto inventory (condensed)

Session toggles · `direction_mode="long_only"` (overrides `SHORT_ALLOWED=True` — `policy/trading_policy.py:26, 48-55`, `immune.py:59`) · `DIRECTION_MIN_SCORE=0.02` · penny bar: sub-\$1 needs `|score| ≥ 0.85` (`PENNY_PRICE=$1.00`, `PENNY_SCORE_BAR=0.85` — `policy/trading_policy.py:57-66`, `immune.py:46-48`) · safety `authorized` snapshot (see §3) · score admission `|score| > dynamics.threshold × patience` · governor 2/cycle + 60 s cooldown · DNN veto `P(win) < 0.52` · realized-EV advisory-only · mechanicals \$5k/\$50/3 positions · portfolio gate (exposure < 1.0, concentration ≤ 0.25) · feed-freshness suppression · 60 s stale-parent sweep · EOD flatten. `MAX_SPREAD_BPS=5` is declared (`immune.py:38`) but the part-1 trace found **no spread veto on the live entry path** (part 3 lists it as an `ib_cycle` execution-path veto without a line ref — treat as unverified).

1.8 Corrections to stale doc claims (verified 2026-09-23)

- `brain/cerebellum.py` **does not exist**; the 27-indicator alpha is `brain/indicators.py:30-55`. Top-level `cerebellum.py` is only the 5-core fallback.
- "0.7·cortex + 0.3·SNN" is wrong: with `NEURO_BLEND_ENABLED=False` the live formula is `0.7 × cortex + 0.3 × 0` — pure damping.
- Entry parent is **LIMIT at mid**, not market (only closes are MKT).
- `_check_safety` / `_exec_decision` do not exist in current `ib_cycle.py`; execution is `_execute_verdict` / `_execute_entries`.
- The realized-EV "gate" never refuses (docstring: "NEVER refuses... only scales").
- `decide_entry` never reads `cortex`'s BUY/SELL verdict — only score/direction feed sizing; the R1 "cortex is sole verdict emitter" claim doesn't match the live admission path.

2. Brain organs — live vs dead

Audit of every learning/adaptive organ against `src/hanoon_prime/` (full table in `part2_organisms.md`).

Organ	LIVE_CHECK (2026-09-18) claimed	Verified 2026-09-23
STDP reward learning	REPAIRED + GATED	GATED OFF — <code>NEURO_LEARN_ENABLED=False</code> (<code>immune.py:153</code>); traces mutate (<code>brain/neurons/bridge.py:148-149</code>) but write- back to scorer weights only when flag true (<code>:150-151</code>)
Neuromorphic score/blend	(implied live)	GATED OFF — <code>NEURO_BLEND_ENABLED=False</code> (<code>immune.py:148</code>); <code>_compute_neuro_score</code> returns 0.0 (<code>orchestrator.py:760-761</code>); feed path still runs every tick (cost without effect)
AttractorMemory	PERSISTED (opt-in)	LIVE — writes on every real close (<code>orchestrator.py:1459-1475</code> → <code>bridge.store_outcome</code>); production persists (<code>ib_adapter.py:47</code> , <code>runtime/attractor_memory.json</code> current). Recall-to-scoring consumers not fully mapped
SleepReplayEngine	REPAIRED + GATED	LIVE (condition-gated, not flag-gated) — constructed <code>orchestrator.py:176-185</code> ; S2 <code>_maybe_sleep_replay</code> (<code>consolidation.py:177-</code> <code>190, 505-518</code>); fires on 30-min idle or session close, ≥3 buffered patterns (<code>SLEEP_MIN_PATTERNS=3</code> — <code>learning_config.py:94</code>). Effect bounded only because neuro gates are off
DynamicThresholdAdapter (neuro)	WIRED + GATED	GATED OFF — <code>NEURO_ADAPTIVE_THRESHOLD_ENABLED=False</code> (<code>immune.py:157</code>); feed runs, then fixed thresholds restored (<code>adaptive_thresholds_wiring.py:30-53</code>)
MoE experts	WIRED + GATED	GATED OFF — <code>NEURO_MOE_GATE_ENABLED=False</code> (<code>immune.py:162</code>); <code>route_experts</code> immediate no-op (<code>moe_gate.py:112-134</code>). Classic cross- asset lead-lag scoring is separately LIVE (<code>orchestrator.py:851-854</code>)

Organ	LIVE_CHECK (2026-09-18) claimed	Verified 2026-09-23
Nine-rule learned exits	LIVE	LIVE — <code>update_from_outcome</code> on every close (<code>orchestrator.py:1533-1556</code>); consumed in exit-ladder Tier 2 (<code>exit_ladder.py:154-165</code>); <code>runtime/adaptive_thresholds.json</code> <code>update_count</code> 2091
ExitPolicy.adapt_from_realized	LIVE ("only live learning")	LIVE (<code>orchestrator.py:1624</code> , <code>brain/exits.py:253-281</code>) — but it is not the only live learning (headline is false)
Episodic memory	(assumed live)	LIVE — <code>episodic.add</code> on close (<code>orchestrator.py:1467-1477</code>); modifier read every tick, ± 0.10 bound (<code>orchestrator.py:952-967</code> , <code>episodic.py:100-108</code>)
Strategy bandit/registry	(Phase H)	LIVE — per-tick Beta-posterior select (<code>strategy_bandit.py:72-107</code>); updates on real + shadow closes; <code>runtime/juli_strategy_bandit.json</code> : 5,787 trials, 23.9M selects, 893k overrides
StrategyResearch / HALIM <code>/v1/research</code>	(Phase H)	LIVE background loop — S2 <code>_maybe_research</code> (<code>consolidation.py:177-190, 203-210</code>), 5-min throttle, no feature flag (<code>strategy_research.py:62-107</code>); POSTs <code>http://127.0.0.1:8765/v1/research</code> ; graceful on failure
HALIM modifier	warming up	LIVE — S2-pollled, 60 s cache (<code>halim_adapter.py:125-133, 213-233</code>); folded into score (<code>orchestrator.py:735-743, 858-863</code>)
Meta DNN gatekeeper	ACTIVE, never trained	LIVE — <code>META_DNN_ENABLED=True</code> (<code>learning_config.py:29-31</code>); <code>_dnn_gate_veto</code> (<code>orchestrator.py:542-544</code>); $9 \rightarrow 32 \rightarrow 16 \rightarrow 1$ MLP, admit if $P(\text{win}) \geq 0.52$ (<code>meta_label_dnn.py:211-237</code>); train report accuracy 0.5896, <code>p_win_median</code> 0.3325 (<code>juli_meta_dnn.report.json</code>); fails open on defective model (see §8)
Nash learning/veto	(implied live)	LIVE — predict every tick (<code>cognitive/nash.py:44-68</code>), penalty ≤ 0.15 in

Organ	LIVE_CHECK (2026-09-18) claimed	Verified 2026-09-23
		<code>_stabilize</code> , outcome recorded per close
Reflector weight adaptation	(learning loop)	LIVE — 2.4:1 loss/reward ratio (<code>reflection.py:44-55, 77-104</code> , <code>brain/config.py:53-54</code>); persists to <code>juli_regime_weights.json</code>
LearnedExitPolicy	(not in audit)	PARTIAL — <code>record()</code> every close (<code>orchestrator.py:1478</code>) but <code>get_policy()</code> has no callers : write-only learning
NetworkStepper	DELETED	DEAD — confirmed gone

Feature-flag inventory (verified values)

From `immune.py` : `DYNAMIC_PRIOR_TOP_ENABLED=True (:81)` · `PROBE_RECOVERY_ENABLED=False (:94)` ·
`CONTRARIAN_MODE_ENABLED=False (:105)` · `CALIBRATION_NUDGE_ENABLED=False (:114)` ·
`HYSTERESIS_EXIT_ENABLED=False (:122)` · `DELIBERATION_TRACE_ENABLED=False (:126)` ·
`HALIM_EVIDENCE_LEARNING=False (:127)` · `NEURO_BLEND_ENABLED=False (:148)` ·
`NEURO_LEARN_ENABLED=False (:153)` · `NEURO_ADAPTIVE_THRESHOLD_ENABLED=False (:157)` ·
`NEURO_MOE_GATE_ENABLED=False (:162)` · `TELEMETRY_AUTH_ENABLED=True (:178)` ·
`ENTRY_REGIME_GATE=False (:196)` · `ENTRY_COST_AVERSE_GATE=False (:197)` ·
`ALLOW_EXTENDED_HOURS=True (:272)` · `FRACDIFF_ENABLED=True (:281)` · `LABEL_ENABLED=True (:293)` ·
`WFA_PURGE_ENABLED=True (:302)`. From `brain/learning_config.py` : `META_DNN_ENABLED=True (:29-31)`. ⚠

The three AFML flags carry "off-by-default" comments but are set **True** — intent unconfirmed (comment/code drift).

Key corrections to LIVE_CHECK.md

1. "Only live learning is ExitPolicy" is materially false — at least 8 other learners run on the close path.
2. Sleep replay is condition-gated and live, not flag-off.
3. Nine-rule "zero callers" is stale — live via exit-ladder Tier 2.
4. Attractor persistence is production-enabled, not merely opt-in.

3. Safety nets

3.1 Hard limits (literals locked by `tests/test_constants_contract.py` ; R6 forbids env configurability)

Limit	Value	Defined	Enforced
Max position notional	\$5,000/trade	<code>immune.py:57</code>	sizing ceiling <code>brain/risk.py:136-166</code>
Max loss per trade	\$50	<code>immune.py:58</code>	sizing ceiling, never raises
Max concurrent positions	3	<code>immune.py:59</code>	<code>risk.py:213-217</code> ($\geq 3 \rightarrow$ zero-size); <code>portfolio_gate.py:69-72</code>
Daily loss $\rightarrow$ ordinary halt	-\$200	<code>immune.py:60</code>	<code>brain/policy/safety.py:76-94</code>
Daily loss $\rightarrow$ latched kill	-\$500	<code>immune.py:61</code>	<code>safety.py:76-94, 131-164</code>
3 consecutive losses $\rightarrow$ pause	3	<code>immune.py:62</code>	<code>safety.py:76-94</code> — indefinite halt; PAUSE_DURATION_MIN=60 (<code>immune.py:63</code>) is never read
Max entries/cycle; reuse cooldown	2; 60 s	<code>immune.py:55-56</code>	<code>policy/governor.py:31-42</code>
EOD flatten	enabled, 5-min window	<code>policy/trading_policy.py:29-35</code>	<code>ib_cycle.py:904-934</code> (MKT, brain skipped)

Strict `<` comparisons: exactly `-$200.00` / `-$500.00` does **not** trip.

3.2 SafetyProducer semantics (`brain/policy/safety.py`)

- Defaults `enabled=False`, `halted=False`, `latched=False` (`safety.py:29-43`); every new `ConsolidationEngine` constructs a fresh disabled producer (`brain/consolidation.py:73`); the live bot explicitly keeps safety off (`ib_adapter.py:47-54`). **Nothing persists state — a restart clears even a latched kill.**
- `authorized()` check order (`safety.py:76-94`): (1) existing latch vetoes even when disabled; (2) **if disabled $\rightarrow$ allow immediately — the `-$500` kill check is skipped unless already latched;** (3) `daily_pnl < -500` $\rightarrow$ latched kill; (4) existing halt $\rightarrow$ veto; (5) `daily_pnl < -200` $\rightarrow$ halt; (6) 3 consec losses $\rightarrow$ halt; (7) `position_count > 3` $\rightarrow$ halt.
- Ordinary halt: sets fields, journals, notifies, returns (`False`, `reason`) — **never stops the process, never closes anything.**

- Kill: sets halted+latched, journals/notifies, invokes the one hook → `executor.cancel_all()` (`ib_adapter.py:95-104`) — **cancels open orders only; does NOT flatten positions.**
- `resume()` clears ordinary halt but not the latch; `release_kill()` re-arms; `set_enabled(False)` clears ordinary halt as a side effect.
- Wiring: slow cortex publishes `policy_state` every ~30 s (`consolidation.py:34-35,111-151`); fast path reads the snapshot in `_apply_fast_gates` (`orchestrator.py:416-458`) — **up to ~30 s of stale authorization**; no synchronous `authorized()` on the hot path.
- `SafetyNetStopped` exception exists (`ib_cycle.py:116-117`) with no live raise found. Sim-path `Hippocampus.check_safety_nets()` raises on the same trips but is constructed with `safety_enabled=False` in live (`ib_adapter.py:47-54`, `orchestrator.py:~150`) — inert.

3.3 Bypass inventory (ways limits get weakened/disabled)

1. `POST /safety-net {"action":"disable"}` — disables producer (kill check skipped unless latched); also clears any ordinary halt (`safety.py:54-58`).
 2. `POST /safety-net {"action":"resume"}` — clears halt immediately (also forces `authorized=True` in the published snapshot — `orchestrator.py:674-683`).
 3. `POST /safety-net {"action":"kill_release"}` — re-arms latched kill (`safety.py:95-101`).
 4. **Process restart** — resets enabled/halted/latched; kill does not survive.
 5. `POST /config` — cannot weaken the immune-literal limits, but **can disable EOD flatten** and enable long-hold horizons (`telemetry.py:559-593`).
 6. Disabled-by-default + check ordering — a -\$500 day with safety off never latches (contradicts `CONTRACT.md` principle 9 "non-overridable").
 7. Kill = cancel-only (no flatten). 8. 30 s stale authorization window. 9. `PROBE_RECOVERY_ENABLED=False` — no bypass today; enabling it would let one gated entry override a halt.
- Auth posture on the POSTs: **disputed between workstreams** — parts 2 and 3 read `TELEMETRY_AUTH_ENABLED=True` in `immune.py:178` (bearer-gated in current code; token at `runtime/telemetry.token`; bind `127.0.0.1`); part 4 read `_authorized()` (`telemetry.py:695-699`) as fail-open when the flag is off and described off as the default. The literal is True per two independent reads; the fail-open *design* (returns True when off/unset) remains a latent risk, and the live tunnel is the real perimeter. Verify at runtime via `GET /auth` behavior.

4. Config and thresholds

- **The live entry threshold is** `Dynamics.threshold` — base `SIGNAL_THRESHOLD=0.58`, adaptive in `[THRESHOLD_MIN=0.45, THRESHOLD_MAX=0.70]` (`brain/config.py:24-27`). It gates entries at `orchestrator.py:1386` (`_maybe_size`), ~636 (`_size_entry`), and seeds `cortex._threshold` each cycle (~1245). Horizon patience scales it: scalp 1.00 → longterm 0.90 (`brain/horizons.py:42-49`).
- **Dashboard "LIVE 0.4500"** = `brain.dynamics.threshold` (`brain/telemetry_facade.py:61`). It drifted to the 0.45 floor via **quintile self-calibration** (`brain/dynamics.py:62-77`: every tick, once

≥ 20 scores accumulate, the bar anchors at the 90th percentile of recent |scores|) — this runs with **zero trades**, so the live bar can sit at the floor while `memory.threshold` (persisted target, `runtime/juli_state.json`, refreshed only on real closes — `orchestrator.py:1518-1530`) still reads 0.58. They are different quantities; both are "correct."

- **Per-close adaptation** (`brain/dynamics.py:84-119`): prediction error $\geq 0.50 \rightarrow +0.01$; $< 0.45 \rightarrow -0.005$; $+0.025$ per losing confidence bin (≥ 10 losses, 0 wins), capped $+0.15/\text{close}$.
- **Dead/confusing constants:** `immune.ENTRY_THRESHOLD=0.65` (`immune.py:30`) — unused on the live path, unasserted by the contract test (legacy). `DECISION_THRESHOLD` (neuromorphic neuron, 0.6) — exported but consumed by nobody (`tests/test_dynamic_threshold_wiring.py:1-7`).
- **No hard confidence gate:** `CONFIDENCE_FLOOR=0.50` only floors reported confidence (`orchestrator.py:~1240`); `_maybe_size` gates on score only.
- **EV gate never refuses:** `ev = (p·r - (1-p)) · direction_mod` (`brain/ev_gate.py:75`) — **no fee drag subtracted**; `round_trip_cost_fraction()` exists (`immune.py:~192-205`) but is wired only via `ENTRY_COST_AVERSE_GATE`, default off (`immune.py:179`). `should_enter=False` paths exist (`ev_gate.py:104-110`, floors `ENTRY_EV_THRESHOLD=0.05` / `CONSERVATIVE_EV_MIN=0.07` — `brain/config.py:175-176`) but `RiskEngine.evaluate()` discards `should_enter` and only scales size ("NEVER refuses... only scales" — `risk.py:190-199`).
- EOD: within 5 min of close `_stabilize` zeroes scores (`orchestrator.py:800-810`); `_check_eod_flatten` (`ib_cycle.py:904-934`) force-closes intraday horizons only.

5. Validation harness

5.1 Walk-forward (`src/hanoon_prime/wfa.py` , 521 lines)

- Per ticker: load `<data_dir>/<TICKER>_1min.csv`; split into `DEFAULT_FOLDS=6` contiguous non-overlapping OOS windows after warmup (`wfa.py:92-112`); simulate with **static production weights** (no fitting $\rightarrow$ no in-fold leakage); a trade counts only if it exits strictly inside `[start, end)` (`wfa.py:262-266`).
- **Pooled universe stats:** `_pooled_returns` concatenates PnL of all trades from **admissible** tickers ($\geq \text{MIN_TRADES}=30$ OOS trades — `wfa.py:43, 278-285`).
- **Deflated Sharpe** (`wfa.py:296-330`): observed pooled Sharpe – 95th percentile of best-by-chance null over `n_trials` bootstraps (`BOOTSTRAP_ITERS=2000`, seeded RNG — `wfa.py:45-46`). $\leq 0 \Rightarrow$ no edge after deflation.
- **PBO** (`wfa.py:345-390`): CSCV over trial half-splits (capped 64); 0.5 if < 4 tickers. **Reported, never gated.**
- **Purge/embargo** helpers exist (`wfa.py:117-157, 185-211`) but the shipped WFA never calls them (no training set); they're used by research harnesses (`train_meta_dnn.py`, phase scripts).

5.2 Reproduce command

`scripts/wfa_report.py` (101 lines; exit 0 only on universe PASS — `:94-101`):

```
python scripts/wfa_report.py --data-dir data/fixtures --tickers ALL --output reports/wfa_abs_new.json
```

reports/wfa_abs_new.json is byte-identical in content to reports/phase2_wfa.json (same fixtures, folds=6). The abs batch naming (written 2026-09-23 14:41-14:44 UTC) is undocumented in-repo. CI wf-gate (.github/workflows/ci.yml:94-102) runs the identical command with no continue-on-error .

5.3 Pass criteria (wfa.py:425-462)

Ticker PASS: ≥ 30 OOS trades **and** `ev_per_trade > 0` **and** per-trade Sharpe > 0 . Universe PASS: ≥ 1 admissible ticker **and** `deflated_edge > 0` **and** `pooled_sharpe > 0`. Related R2 gate: `python -m hanoon_prime.backtest + scripts/check_profit_gate.py` (exit 1 if any ticker $EV \leq 0$ — `backtest.py:115-136`). **Latest result (2026-09-23): FAIL** — pooled Sharpe -0.0456 , deflated edge -0.3068 (126 trials), PBO 0.4844; 4/23 tickers admissible, MARA the only PASS, 19 INSUFFICIENT. **Protocol drift:** `task_plan.md` Phase 4 pre-locks `deflated > 0.05 AND PBO < 0.05`, but `scripts/paper_run.py:176-184` reuses `wfa.verdicts` verbatim — the stricter criteria are **not enforced in code**. Also `_count_trials` (`wfa.py:332-338`) counts ticker $\times$ fold cells over **all** tickers (126) while pooling only **admissible** ones (4) — deflation penalty calibrated to far more trials than feed the Sharpe (conservative bias on thin universes).

6. Telemetry / dashboard (src/hanoon_prime/telemetry.py , 1774 lines)

- `ThreadingHTTPServer` on `127.0.0.1:TELEMETRY_PORT` (`telemetry.py:~1699`); refresher rebuilds snapshot every 1 s (`SNAPSHOT_INTERVAL=1.0` — `:51`); dashboard (`hanoonweb.xyz` via cloudflared tunnel) polls `GET /snapshot` or holds `GET /stream` SSE (`SseRegistry.broadcast`).
- **29 GET endpoints**, none authenticated: `/snapshot`, `/stream`, `/health`, `/journal`, `/positions`, `/safety-net`, `/brain`, `/trades`, `/system2`, `/pipeline`, `/risk`, `/config`, `/halim`, `/verdicts`, `/session`, `/account`, `/resources`, `/inspection`, `/logs`, `/ib` (raw broker surface: quotes, orders, fills, executions, IB errors), `/decisions`, `/trade-quality`, `/trust`, `/vitals`, `/exec-quality`, `/monitors`, `/metrics`, `/auth`.
- **State-changing POSTs** — gated by `_authorized()` (`telemetry.py:695-699`), which **returns True when TELEMETRY_AUTH_ENABLED is off/unset** (fail-open design; current literal reads True — see §3.3 dispute):
 - `POST /safety-net` — `enable` / `disable` / `resume` / `kill_release` (re-arms the latched kill — `:493-510`).
 - `POST /config` — `sessions`, `direction_mode` (both/long_only/short_only), EOD flatten, horizons (`:559-593`).
 - `POST /flatten` — sets `_FLATTEN_REQUESTED` → main cycle flattens **all positions** (`:1426-1460`). **Missing from POST_ROUTES** (`:78`) though present in `POST_HANDLERS` (`:83-87`) — works but invisible to allow-list audits.

- `GET /auth` **returns the bearer token** to CORS-allowed browser origins (:700-725); the allow-list auto-includes the live tunnel URL from `runtime/tunnel_url.txt` (:657-668).
- Design note in code: "tunnel is the perimeter" (:704) — i.e., public exposure of the tunnel = public exposure of all of the above.

7. Known bug history (`FIXES.md` , 264 lines, append-only, guard-enforced)

Class	Pattern	Entries
E — unwired capability assumed working	dead/dormant features treated as live	6 (most recurrent; latest 2026-09-08-07): regime fallback throttle, StrategyGenome zero callers, portfolio-risk stub, <code>_watched</code> lazy init, penny-score bar, direction default
B — single-writer violations	two modules writing one store	4: trade double-counting, re-adopt mid-flatten, <code>cancelAllOrders</code> clobbering flatten orders
A — path arithmetic / env assumptions	wrong <code>parents[N]</code> , CWD-relative paths	3
D — silent exception swallowing	systemic failure logged away	3: incl. blocking seeding → skipped protection
C — typed-container truthiness	<code>arr</code> or fallback on numpy → <code>ValueError</code> / silent starvation	1 — but has a permanent static watchdog in the enforcer test

Most instructive:

1. **FIX-2026-09-07-05 (C+D)** — `arr` or fallback on numpy arrays in `juli_feed` killed **every** live entry evaluation (`thinks=+0` , `decisions=0`); invisible to list-shaped unit tests. Fix: explicit `is None / len()` checks + snapshot-shaped fixtures. Lesson: fixtures must be numpy-shaped, not convenient lists.
2. **FIX-2026-09-07-07 (E+B)** — portfolio risk manager was a stub (`update(net_liq, {})`) — positions hardcoded empty; exposure/concentration logic could never fire. Full port + wiring + `/risk` endpoint.
3. **FIX-2026-09-07-02 (B)** — every real trade double-counted in learning (orchestrator + Reflector both writing). Fix: documented single-writer ownership + exact-delta smoke asserts. Standing defenses: `scripts/smoke_live.py` (23 checks), `PipelineMonitor` , journal-enforcer test, deliberately-dormant monitor list.

8. Top 10 concrete problems (ranked by safety/correctness impact, then edge impact)

1. Safety net off by default; the -\$500 kill is skipped while disabled; the latch doesn't survive

restart. `brain/policy/safety.py:29-43` (defaults), `:78-84` (disabled early-return precedes the kill check), `brain/consolidation.py:73` (fresh disabled producer per engine). A -\$500 day with safety off never latches; a restart wipes even a latched kill. *Fix:* evaluate the kill condition before the disabled early-return; persist `{enabled, halted, latched}` to `runtime/safety_state.json` and restore on boot; default `enabled=True`. — **Safety: critical.**

2. Kill cancels orders but never flattens positions.

`safety.py:131-164` → `executor.cancel_all()` (`ib_adapter.py:95-104`). After a kill day the bot holds its losers into the next session with entries vetoed but nothing forcing a flat book. *Fix:* second kill hook → `executor.close_all_positions(...)` (market) before/after cancel-all; journal it. — **Safety: high.**

3. Fixed 0.7 score damping with the neuromorphic blend disabled.

`brain/orchestrator.py:1285-1296`; `NEURO_BLEND_ENABLED=False` (`immune.py:148`). Every cortex score is multiplied by 0.7 for zero neuro contribution: the penny bar (0.85) becomes nearly unreachable, and the declared threshold band [0.45, 0.70] no longer means what it says in cortex units (the quintile self-calibration absorbs the distortion). *Fix:* when the gate is off, `blended = base.score + cal_adj`. — **Correctness: high (warps every entry decision).**

4. Sleep replay rewards losers.

`brain/neurons/sleep.py:167-171` sets `reward = 1.0` if `weight >= 1.0` else `0.5`, but the scheduler tags losers weight 3.0 and winners 1.0 (`brain/sleep_scheduler.py:51-73`) — both satisfy `>= 1.0`, so losers get the positive reward the docstring says they shouldn't. *Fix:* thread explicit `(pattern, won)` polarity into `_replay_pattern`. — **Edge: high (actively teaches the wrong lesson).**

5. Meta DNN gatekeeper fails open.

`MetaDNN.infer` returns `(True, META_WIN_THRESHOLD, scale)` on defective/unbuilt model (`brain/meta_label_dnn.py:175-210`); `MetaLabelModel._fallback_gate` always admits (`brain/meta_label.py:186-199`); `gate()` falls back on any DNN exception (`:142-156`). A missing/corrupt `juli_meta_dnn.json` silently disables the entry veto. *Fix:* fail closed (or startup warning) when `live_snapshot()["bypassed"]`. — **Safety/correctness: high.**

6. The EV and confidence "gates" never refuse; no fee drag in EV math.

`RiskEngine.evaluate()` discards `should_enter`, only scales size 0.5–1.0 ("NEVER refuses... only scales" — `brain/risk.py:190-205`); no hard confidence gate exists; `ev = (p·r - (1-p)) · direction_mod` subtracts no costs (`brain/ev_gate.py:75`), and `round_trip_cost_fraction()` is wired only via the default-off `ENTRY_COST_AVERSE_GATE` (`immune.py:179`). The contract's central promise — no trade without positive expectancy — is advisory-only in code. *Fix:* make `should_enter=False` refuse (or document the downgrade); subtract fee+slippage in the refuse math. — **Edge: high.**

7. Entry parent is a limit at mid, not a market order; governor cooldown stamped at placement.

`ib_executor.py:134`; `ib_cycle.py:716`; stale-parent sweep at `:262-298`. Entries can rest unfilled in a fast move, then get swept after 60 s; re-entry needs a fresh ENTER **plus** the 60 s cooldown that was stamped even though nothing filled — the brain believes it's "in" while flat. *Fix:* marketable limit offset

(or MKT parent + STP/LMT children); stamp cooldown on confirmed fill (`_confirm_fill`). —

Correctness: medium-high.

8. ATR trailing-stop logic is dead code. `IBExecutor.monitor_orders` / `_handle_parent`

(`ib_executor.py:419-494`) implement trailing, but no call site exists in `ib_cycle.py` or `ib_adapter.py`; `exits.py` docstrings still claim it runs alongside mechanical exits. *Fix:* call `monitor_orders` per cycle (with skip-if-closing) or delete the code and its claims. — **Correctness: medium.**

9. Validation-harness drift: Phase-4 criteria unenforced; WFA trial-count mismatch.

`scripts/paper_run.py:176-184` reuses `wfa.verdicts` verbatim, ignoring the pre-locked `deflated > 0.05 AND PBO < 0.05` (`task_plan.md` Phase 4); `_count_trials` (`wfa.py:332-338`) counts cells over all tickers (126) while pooling only admissible ones (4); shipped WFA never calls purge/embargo (`wfa.py:117-157`). The harness cannot currently certify edge the way the protocol requires. *Fix:* implement the P5 check in `run_paper`; count trials within the admissible set; document the conservatism. — **Edge-measurement integrity: medium.**

10. Telemetry mutation surface is one misconfiguration from unauthenticated remote control.

Fail-open `_authorized()` (`telemetry.py:695-699`); `GET /auth` serves the bearer token to CORS origins including the auto-added tunnel URL (`:657-672, 700-725`); `POST /flatten` (real orders) missing from `POST_ROUTES` (`:78` vs `:83-87`); loopback bind + cloudflared tunnel = public perimeter. Current `TELEMETRY_AUTH_ENABLED=True` literal (`immune.py:178` , two reads) mitigates today, but the design fails open. *Fix:* default auth on and verify at runtime; never serve the token to browser origins; single source of truth for POST routes. — **Security/safety: medium (latent critical).**

Honorable mentions: 30 s stale authorization window on the entry hot path (`consolidation.py:34-35` → `orchestrator.py:416-458`); documented 60-min consecutive-loss pause unimplemented — halt is indefinite (`immune.py:63` never read); HALIM threshold recommendations crash on the read-only `Dynamics.threshold` property (`halim_recommendations.py:113-118` vs `brain/dynamics.py:79-82`); held tickers consume the 2-per-cycle entry budget (`juli.py:76-106` → `ib_cycle.py:702-706` "SKIP open"); `LearnedExitPolicy.get_policy()` write-only; AFML "off-by-default" comments contradict `True` values (`immune.py:281, 293, 302`); dead `ENTRY_THRESHOLD` / `DECISION_THRESHOLD` constants invite misreading.

Appendix — part files and key entry points

- `part1_live_path.md` — full tick → fill trace, 23-gate table, 5 candidate problems.
- `part2_organs.md` — organ table, flag inventory, LIVE_CHECK corrections, 5 candidate problems.
- `part3_safety.md` — limit table, SafetyProducer semantics, threshold analysis, 12-item bypass inventory, 5 candidate problems.
- `part4_validation.md` — WFA mechanics, reproduce command, 33-endpoint inventory, FIXES.md analysis, 51-script inventory, 5 candidate problems.

- Live entry: `src/hanoon_prime/cli.py:85` → `ib_adapter.py:160` (`_cycle`) → `ib_cycle.py:561` (`_run_brain_cycle`) → `juli.py:44` (`tick`) → `brain/orchestrator.py:263` (`decide_entry`) → `ib_cycle.py:645` (`_execute_entries`) → `ib_executor.py:62` (`place_bracket`).
- Validation: `python scripts/wfa_report.py --data-dir data/fixtures --tickers ALL --output reports/wfa_abs_new.json` (exit 0 ⇔ universe PASS).